

# RONIT RAJ BAISHYA

Guwahati, Assam, India | ronitb@vt.edu | (540) 557-8814 | linkedin.com/in/ronitrajbaishya

## RESEARCH PROFILE

Interests focus on the application of scientific machine learning and computational science for physical systems, mainly emphasis on neural operators, physics-informed neural networks for structural elements and dynamical systems, modeling, analysis, inverse problems, uncertainty quantification, and parameter/hidden state estimation. Broader interests in PDE-constrained optimization, design, estimation, and control. Experience in the application of machine learning, signal processing, statistical modeling, modal analysis, structural engineering, and dynamical systems.

## EDUCATION

|                                                                                                                             |                                              |
|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>Lehigh University</b><br>PhD in Structural Engineering                                                                   | Bethlehem, Pennsylvania<br>Aug 2025–Aug 2026 |
| <b>Virginia Polytechnic Institute and State University</b><br>MS in Civil Engineering, Structural Engineering and Materials | Blacksburg, Virginia<br>May 2024             |
| <b>Jorhat Engineering College</b><br>Bachelor of Engineering in Civil Engineering                                           | Jorhat, Assam, India<br>Sept 2021            |

## RELEVANT COURSES

|                                        |                                                                                                                                                                                                                                                          |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ML/AI</b>                           | Scientific Machine Learning, Data Mining, Statistics and Machine Learning, Machine Learning-I, Data Analytics, Time Series Analysis                                                                                                                      |
| <b>System Dynamics</b>                 | Nonlinear Systems and Chaos, Instrumentation and Signal Processing, Applied Linear Controls, Structural Dynamics                                                                                                                                         |
| <b>Civil and Structural Background</b> | Structural Mechanics, Structural Analysis, Computer Methods of Structural Analysis, Mechanics of Composite Materials, Intermediate RC and Steel Design, Design of Prestressed Concrete, Fluid Mechanics, Hydrological Machines, Geotechnical Engineering |

## TECHNICAL SKILLS

|                    |                                                                                                                                                                                                                                                                                        |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b> | Python, R, Jupyter notebooks, VS-Code, MATLAB, Mathematica, L <sup>A</sup> T <sub>E</sub> X                                                                                                                                                                                            |
| <b>ML</b>          | Supervised learning, unsupervised learning, PINN, neural operators, CNN, inverse problem, FNO, DeepONet, scikit-learn, pandas, NumPy, JAX, linear regression, classification, clustering, uncertainty quantification, benchmarking, hyperparameter tuning, model evaluation, inference |
| <b>Computing</b>   | Time-series analysis, ARIMA, ACF/PACF, signal processing, FFT/DFT, feature engineering, statistical modeling, Kalman filtering, modal analysis                                                                                                                                         |
| <b>Deployment</b>  | Streamlit dashboards, Git/GitHub, reproducible scripts, technical documentation, visualization pipelines                                                                                                                                                                               |
| <b>Engineering</b> | Abaqus, MathCAD, MIDAS, Revu Bluebeam, MicroStation, SAP 2000, ETABS, RISA-3D, STAAD.Pro, Response 2000, MS PowerPoint, MS Excel                                                                                                                                                       |

## PUBLICATIONS

|                                                                                                                                                                                                        |          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Predicting Mechanical Properties of Glass-Epoxy Composites Using Supervised Learning</b><br><i>Journal of Reinforced Plastics and Composites</i> , Sage Journals<br>DOI: 10.31224/07316844251318775 | Feb 2025 |
| <b>India Flood Inventory: Creation of a Multi-Source National Geospatial Database to Facilitate Comprehensive Flood Research</b><br><i>Natural Hazards</i> , Springer                                  | May 2021 |

## PREPRINTS

|                                                                                                                                                                                                                                |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Modal Identification and Environmental Effects on Natural Frequencies Using openLAB Research Bridge</b><br>EngrXiv Archives (Submitted: Journal of Advances in Structural Engineering, Sage Journals)<br>DOI: 10.31224/8196 | Sept 2026 |
| <b>Gas Stove Activity Detection Using Raspberry Pi and Pi Camera: A Signal Processing Approach</b><br>TechRxiv Archives<br>DOI: 10.36227/techrxiv.172556906.62692261/v1                                                        | Sept 2024 |
| <b>Review of Bridge Structural Inspection Aided by Augmented Reality</b>                                                                                                                                                       |           |

## TECHNICAL PROJECTS

### Physics-Informed DeepONet for Uncertainty-Aware Composite Identification Sept 2026-Present

- Develop a validated 3D orthotropic glass-epoxy three-point-bending FEM model and use it to train a physics-informed DeepONet that reconstructs the structural response from material parameters.
- Then solve the sparse inverse problem to estimate key properties such as  $E_2, G_{12}, G_{13}$ , quantify their uncertainty, and determine which property most strongly governs the observed response.

### Bathymetry-Conditioned CFO for 2D Shallow-Water Dynamics Aug 2026-Present

- So far, the CFO was extended to full-state 2D shallow-water dynamics  $[h, hu, hv]$  using an FNO backbone, added an SWE physics loss, validated, and shown that PI-CFO improves physical consistency and mass conservation at a small cost in trajectory accuracy.
- Next, varying bathymetry  $b(x, y)$  will be added, make the model terrain-conditioned and well-balanced, test unseen terrains and lake-at-rest preservation, and then study resolution transfer such as  $32 \times 32 \rightarrow 64 \times 64$ .

### Environment-Compensated Structural Health Monitoring Streamlit App Mar 2026-May 2026

- Built a reproducible ML pipeline and Streamlit dashboard using openLAB bridge data to model temperature and solar-radiation effects in vibration-based monitoring.
- Implemented OLS, Ridge, and Lasso residual models to generate an Environment-Normalized Damage/Novelty Index for anomaly detection and false-alarm reduction.

### Comparative Deep Learning Models for Image Classification May 2026

- Built and benchmarked MLP, CNN, and Vision Transformer models on MNIST and CIFAR-10 using PyTorch.
- Developed preprocessing, training, inference, ROC/F1 evaluation, and hyperparameter-tuning workflows for reproducible model comparison.

### Clustering Analysis of Simulated and Gene-Expression Datasets Mar 2026

- Applied unsupervised learning to simulated and high-dimensional scientific datasets for pattern discovery and cluster interpretation.
- Evaluated clustering behavior under sparse-label conditions, building methods relevant to environmental monitoring, hydrologic systems, and sensor-driven anomaly detection.

### Physics-Informed Neural Networks for openLAB Bridge Damage Detection Oct 2025-Dec 2025

- Developed a physics-informed ML framework using vibration data, modal-frequency features, and damage indices for structural damage detection.
- Benchmarked the PINN approach against Random Forest, Ridge Regression, and Gradient Boosting to evaluate accuracy, interpretability, and generalization.
- Connected physical constraints with data-driven inference to support reliable ML models for civil and mechanical systems with limited labeled data.

### Forecasting and Trend Analysis of Wind Speed and Wind-Power Generation Apr 2024-May 2024

- Modeled wind-speed and power-generation trends using ACF, PACF, cross-correlation, and ARIMA-based time-series forecasting.
- Developed forecasting skills transferable to environmental systems, water-resource modeling, renewable-energy planning, and climate-sensitive infrastructure.

### Periodicity and Nonlinear Behavior of Lorenz Attractors Apr 2024-May 2024

- Analyzed Lorenz-system behavior under varying  $\rho$  values to study nonlinear dynamics, sensitivity, and chaos in physical systems.

### Development of an Output Feedback Controller Apr 2024-May 2024

- Designed a discrete-time LQR output-feedback controller with Kalman filtering for state estimation under incomplete measurements.

## RESEARCH, AND TEACHING EXPERIENCE

|                                          |                                  |
|------------------------------------------|----------------------------------|
| Indian Institute of Technology, Guwahati | Guwahati, Assam                  |
| <i>Research Intern</i>                   | August 2026-July 2027 (expected) |
| Lehigh University                        | Bethlehem, Pennsylvania          |
| <i>Graduate Research Assistant</i>       | May 2026-Aug 2026                |
| Lehigh University                        | Bethlehem, Pennsylvania          |
| <i>Graduate Teaching Assistant</i>       | Jan 2026-May 2026                |
| Virginia Tech                            | Blacksburg, Virginia             |
| <i>Teaching Assistant</i>                | Dec 2023-May 2024                |

**Virginia Tech Vibration Testing Laboratory**  
*Graduate Assistant*  
**HydroSense Lab, Indian Institute of Technology, Delhi**  
*Research Intern*

Blacksburg, Virginia  
Sept 2022–Dec 2022  
Delhi, India  
July 2020–May 2021

**INDUSTRY EXPERIENCE**

---

**Kiewit Corporation**  
*Structural Engineer*

Lone Tree, Colorado  
June 2024–Apr 2025

**Kiewit Corporation**  
*Structural Engineer Intern*

Lone Tree, Colorado  
May 2023–Aug 2023

**Architects Collaboratives**  
*Graduate Engineer Trainee*

Assam, India  
Aug 2021–July 2022

**AWARDS AND FELLOWSHIPS**

---

**Kanai and Bibha Mukherjee Opportunity Fund**, Lehigh University  
**Rossin College Fellowship**, Lehigh University  
**Gibson Fellowship**, Lehigh University

May 2025  
June 2025  
Nov 2025